

Names: Aayush Pathak.
Enrollment: 2405112120135
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Div: B
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What are the potential benefits and drawbacks of implementing Google Duplex for customer service in this context?

* Benefits *

Saves time and costs by automating routine tasks (e.g., reservations, FAQs).
Offer 24/7 availability and faster responses.
Ensures consistent service quality.

* Drawbacks *

May struggle with complex or emotional conversations.
Can make customers uncomfortable if they don't know it's AI.
Risk of technical errors or system failures.

How should the business address ethical concerns related to using AI in customer interactions?

Be Transparent: Let customers know they're speaking to AI.
Offer Choice: Allow them to talk to a human if preferred.

Protect Data: Ensure privacy and compliance with laws.

Maintain Oversight: Keep humans in the loop for complex cases.

Q.3) What measures can be taken to ensure that employees are not adversely affected by this technological adoption?

→ Train Staff: Upskill employees for new, more valuable roles.

Reassure Jobs: Clarify that AI is for support, not replacement.

Involve Them: Get input from staff during AI implementation.

Use Hybrid Approach: Let AI handle simple tasks, humans manage complex ones.

Q.4) In what scenarios might rote learning be more effective than learning by taking advice, and vice versa?

→ Rote learning is useful in early learning stages or exam prep where quick recall is needed.

→ Rote learning is best for memorizing facts, formulas, vocabulary, or procedures (e.g., multiplication tables, periodic table).

→ Learning by taking advice is better for critical thinking, problem-solving, creativity, and applying concepts in real life.

Q.5 How can educators balance these two approaches to maximize student understanding and retention?

- **Combine Methods:** Use rote learning for foundational knowledge, then shift to advice-based learning for applying and analyzing it.
- **Differentiate Instruction:** Adjust the balance based on subject, age, and student needs.
- **Encourage Discussion:** Pair memorization with group work, discussions, and projects to deepen retention.

Q.6 What role can technology play in facilitating these different learning methods?

- **Rote support:** Apps and tools like flashcards (e.g., Quizlet), spaced repetition, and drill games.
- Advice-Based Support:** Online discussion forums, mentorship platforms, AI tutoring, and collaboration tools (like Google Classroom or Padlet).

Q.7 How can the integration of AI tools like this impact the problem-solving process in a corporate setting?

- **Faster Solutions:** AI can analyze large datasets quickly to suggest potential solutions.
- **Data-Driven Decisions:** AI supports evidence-based thinking, reducing guesswork.
- **Enhanced Collaboration:** Teams can focus on creative and strategic tasks while AI handles routine analysis.
- **Continuous Learning:** AI learns from new inputs, improving over time.

Q-8. What are the limitations of relying on AI for problem-solving, and how can they be mitigated?

* Limitations *

- AI may lack context or creativity.
- Biased or outdated data can lead to flawed suggestions.
- Over reliance may reduce human initiative.

* Mitigations *

- Keep humans in decision loops.
- Regularly update and audit AI data.
- Use AI as a guide not a final decision-maker.

Q-9] How can companies ensure that employees continue to develop their problem-solving skills alongside using AI tools?

- Training Programs: Teach companies how AI works and how to interpret its outputs.
- Blended Problem-Solving: Encourage teams to use AI as a support, while solving parts manually.
- Promote Critical-Thinking: Challenge employees to question and improve AI-generated solutions.

Q-10] How effective is learning from examples compared to traditional teaching methods?

- Highly effective for many learners:
- Examples make abstract concepts concrete and easier to understand.
- Improves engagement: Real-life or relatable scenarios keep students more interested.

Supports Deeper Learning: It helps learners understand application, not just theory.

What are the potential biases that AI might introduce in selecting examples, and how can they be addressed?

* Biases AI might Introduce *

Selecting examples that reflect cultural, gender, or socioeconomic biases.

Favoring common patterns while ignoring edge cases or diverse learning styles.

How to address:

Train AI on diverse, representative datasets.

Include human oversight in content selection.

Continuously audit and update content for fairness and inclusivity.

How can this approach be integrated into traditional classroom settings?

Blended Learning: Combine teacher-led instruction with AI-driven examples.

Interactive Platforms: Use smart tools to suggest examples based on student progress.

Teacher Support: Let AI recommend examples, but allow teachers to adjust or enhance them.

Project Based Learning: Encourage students to explore AI-generated examples and create their own.

Q:13) What are the strengths and limitations of inductive learning in making predictions in healthcare?

→ *** Strengths:**

- **Pattern Discovery:** Learns from large datasets to find trends not easily visible to humans.
- **Predictive Power:** Can forecast patient outcomes and suggest treatments based on past cases.
- **Scalability:** Handles vast and complex medical data efficiently.

*** Limitations:**

- **Data Quality Dependency:** Poor or biased data leads to inaccurate predictions.
- **Generalization Risk:** May fail with rare conditions or outliers not seen in training data.

Q:14) How can healthcare professionals validate and trust the recommendations generated by such AI systems?

- **Clinical Oversight:** Always involve doctors in reviewing and approving AI suggestions.
- **Transparent Models:** Use interpretable models or tools (e.g. SHAP, Lime) to explain predictions.
- **Rigorous Testing:** Validate models with diverse, real-world data before deployment.
- **Ongoing Monitoring:** Continuously track accuracy and outcomes over time.

15) What ethical considerations arise when using AI-driven predictions in critical areas like healthcare?

- 1) Patient Consent: Clearly informs patients when AI is involved in their care decisions.
- 2) Bias and Fairness: Ensure the system doesn't reinforce inequalities or exclude minorities.
- 3) Accountability: Define who is responsible for decisions - AI developers or healthcare providers?
- 4) Privacy: Protect patient data with strong security and compliance (e.g., HIPAA, GDPR).